Independent observations, provisional association, and limits to reproducing the Campi Flegrei catalog

Reproduction assessment of Tan et al. (2025)

9 October 2026

Abstract

Tan et al. describe an observational earthquake-catalog workflow, not a coupled physical forward model. This resumed investigation processes original waveforms, applies the released retrained PhaseNet model, runs provisional GaMMA association with an explicitly different external starting profile, and attempts response-corrected relative-magnitude measurements. It also recovers original leveling and GNSS observations. No publication panel is fully independently reproduced. Figure 1B and supplementary S1, S2 and S15C are partial; external routine-catalog context is shown separately. The complete multiyear acquisition, calibrated locations, double-difference relocation, magnitude distribution and focal mechanisms remain incomplete. A successful rebuild establishes execution of the reported limited calculations, not reproduction of unexecuted stages.

1 Scientific specification and input verification

The local `tanmodel.pdf` and `tanmodelsupp.pdf` were extracted before changing outputs. Main captions 1–4, supplementary captions S1–S15 and methods S2–S11 are readable. Two font-weight warnings do not prevent main-text extraction. Both supplied ZIPs were listed and CRC-tested; no file from them was executed. Their inventory and hashes are in `docs/resumed_source_check.json` and `docs/file_inventory.json`.

File	Bytes	Inspection
`tanmodel.pdf`	2140554	Text and captions readable
`tanmodelsupp.pdf`	62164947	Text and captions readable
`science.adw9038_data_s1_to_s3.zip`	12960255	Directory readable; CRC passed; not executed
`science.adw9038_interactive_plot_s1.zip`	8852485	Directory readable; CRC passed; not executed

The data archive contains Data S1 (final event catalog), S2 (associated phases) and S3 (focal mechanisms), plus binary macOS packaging metadata. The other archive contains an HTML seismicity visualization. These are derived products, not original waveform inputs. Data S1 was read only after independent outputs existed, for a separate final comparison of temporal coincidences and horizontal differences. Data S3 was read only for a final row-count audit: the CSV has 74 rows and 58 nonempty gCAP solutions, consistent with the methods. Its caption's 55-event description is inconsistent with the actual file. No supplied solution entered inference, initialization, filtering or magnitude estimation.

The latest instruction permits an external input genuinely required by the published method. Accordingly, routine INGV duration magnitudes are permitted only after independent detection and association, as supplementary S6's explicit magnitude regularization reference. Their use no longer presents the blanket policy conflict stated in the previous report. The picking/association program never opens that

catalog. Historical catalog plots are explicitly external context, not independently detected earthquakes.

The authors' fitted 1D velocity model, station corrections, picks, associated catalog, focal solutions and study scripts remain excluded. Generic PhaseNet/SeisBench, GaMMA and VELEST are attributed method software. The trained PhaseNet artifact is used for new inference; retraining is not claimed. All numerical figures read saved CSV or NPZ outputs. No figure was digitized, and no published image substitutes for an output.

2 Data access and actual coverage

The frozen rebuild snapshot contains 230 external files (435,600,234 bytes). Original and extended processing read 171 waveform files from 26 stations, representing 428.983 channel-hours after unioning overlapping intervals. This is a lower bound on retrieved coverage, not a full-network availability estimate. The study interval is 21 January 2022 through 20 March 2025. All 26 full-period IV/2I station requests returned HTTP 413, explicitly citing a one-billion-sample limit. This is a request-size constraint, not evidence that the archive is unavailable. Half-hour requests were then executed for calendar-selected windows on 21 January 2022, 1 January 2023, 1 January 2024 and 20 March 2025. These windows were chosen independently of every earthquake catalog. Daily requests were also attempted; most timed out after 180 seconds, while two returned complete files. Failed transfers are not counted as full-day coverage. A resumable retrieval tool and exact input manifest accompany the project. Full-period download and processing remain unfinished.

Complete response metadata was obtained for 26 station names in networks IV and 2I. This extends the previous four-station response inventory and includes four temporary-network stations. Metadata epochs do not prove continuous recording. The coverage figure uses returned miniSEED segments and preserves gaps. Coverage totals count each channel separately; they do not imply three-component coverage throughout every station interval. Some single-channel records do not support the three-component picker. Actual input files, channel segments, sampling rates, and unioned intervals are saved in the manifests and coverage tables.

The INGV landing page fails local TLS issuer verification, whereas the official FDSN service works with normal verification. RAN's public pages are readable but waveform downloads require registration/login; credentials were unavailable. No account was created. Individual HTTP 204, HTTP 502 and timeout results remain explicit in the receipt files.

The temporary/local ZIP's directory was read using HTTP byte ranges. It lists 319,890 members and approximately 3.845 GB of uncompressed content. Its actual filename station inventory includes CFB1, CFB3, CROS, V0102, V0105, CRUS and CMDP; it does not establish every station advertised on the landing page. One miniSEED member was read successfully, with 6001 samples over 60 seconds. These observations are cut around author-detected events and therefore cannot supply unbiased continuous event discovery. They were not used to seed analysis. The four 2I stations were instead accessed through continuous FDSN requests.

3 Independent processing and explicit departures

3.1 Picking and provisional association

The picker uses released tensors with a strict tensor-only load, 100 Hz sampling, nominal 30-second windows, ENZ components, standard-deviation normalization, 1500-sample overlap, 250-sample edge blinding and a 0.3 P/S threshold. The release README specifies a 1 Hz highpass; four poles and a causal implementation are explicit choices. Extended processing uses 30-minute cores with 15-second context where recorded. Missing intervals are not interpolated. Repeated picks within 0.25 seconds

at the same station and phase are merged by retaining the larger score. This merging tolerance is an assumption. S picks at CASE, CFB1 and CFB3 are excluded.

The numerical starting profile cited by the paper is Calò and Tramelli (2018). Its retrieved supplement presents that profile graphically; no numerical table was found, and its data statement offers data on request. No plot was digitized. The released fixed 3D file is identified in the model repository as the TomoDD input; the paper attributes its fixed background to Battaglia et al. (2008). Its original-release byte identity remains unverified. The file contains a $23 \times 14 \times 27$ grid of V_P and V_P/V_S; V_S is calculated by division before averaging.

Two independent horizontal averages were computed: 14.02–14.20 degrees E, 40.77–40.86 degrees N, and a central box at 14.11–14.16 degrees E, 40.81–40.84 degrees N. These polygons are explicit assumptions, not inferred figure borders. Bounding nodes and the deep 200 km padding level are omitted from the association profile. Above sea level, the surface velocity is extended to station elevations rather than averaging air nodes. This profile is a sensitivity alternative; the associated events cannot be called an exact reproduction of the paper's first locations.

GaMMA uses its 1D eikonal travel-time option with 0.1 km grid spacing. The association requires at least six picks, without separate P/S minima. Additional declared settings are a 10-second DBSCAN neighborhood, three minimum DBSCAN samples, fivefold mixture oversampling, a time-only likelihood, 1 second-squared covariance prior, horizontal bounds of ±12 km and depth bounds of 0–8 km. Amplitude is omitted from association because the paper does not specify its association-amplitude configuration; the GaMMA placeholder magnitude is removed from saved events. These choices are in `config/association.json` and were not tuned against a supplied catalog.

3.2 Location quality and magnitude attempt

Travel-time residuals are recomputed from each provisional origin and the same eikonal model. I calculate

$$R = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(t_k - t_0 - T_k)^2},$$

and the largest unoccupied azimuth between distinct stations. The generic VELEST guidance requires more than six observations and a gap below 180 degrees for calibration candidates. This is a geometry screen, not acceptance of their waveform interpretations. The paper's 0.25-second rejection is a post-location criterion and is not imposed prematurely as a calibration prerequisite.

The generic VELEST source builds after three platform-only repairs: a FORMAT trailing comma, the CPU clock, and the date/time routine. Its scientific inversion was not run. The small provisional candidate set does not constrain the chosen 21-level P/S model, hypocenters and station corrections. The paper used 373 calibration events. No author-fitted model fills this gap. Consequently, the prescribed VELEST residual-rejection sequence, two TomoDD rounds and regional HypoDD SVD uncertainty tests remain unexecuted.

The magnitude attempt nevertheless measures available waveforms independently for all provisional events, without restricting it to the VELEST calibration subset. A recorded P pick defines the arrival, with the calculated P arrival used where only S is assigned. Full response removal produces velocity with a 0.1, 0.2, 35, 45 Hz prefilter and 60 dB water level; these are stated measurement assumptions, not fully specified publication settings. Signal and noise powers use five seconds after and two seconds before the arrival:

$$\mathrm{SNR} = 10\log_{10}(P_{\mathrm{signal}}/P_{\mathrm{noise}}).$$

Observations below 5 dB are rejected. Candidate event pairs require at least two common stations and separation below one-fifth of both source–station distances. The measured equation is

$$M_i - M_j = \log_{10}(A_{is}/A_{js}).$$

The implemented least-squares objective weights channel equations by squared waveform correlation and uses routine M_d priors with declared unit regularization weight. A connected component must contain measured amplitude-difference equations and a matched external anchor. It cannot acquire a magnitude merely because an external M_d exists. This run produced no qualifying edge and therefore no calibrated magnitude. Completeness and b-value cannot be estimated from an empty independently calibrated sample.

3.3 Hybrid waveforms and search

The three templates use the times written in supplementary S9. Matching uses eight-second CAAM/COLB vertical windows, a causal four-pole 0.2–6 Hz filter, mean two-station normalized correlation above 0.25, eight-second peak separation and exclusion of ten-second segment edges. Every supported downloaded common interval is scanned, with duplicate exposure removed. This is not a three-year search.

Candidate frequency indices use separate response-corrected, eight-second CAAM windows:

$$\mathrm{FI} = \log_{10}(P_{8\text{–}15\,\mathrm{Hz}}/P_{2\text{–}5\,\mathrm{Hz}}).$$

The matching bandpass is not reused for FI. Candidate waveform panels contain largely persistent broadband fluctuations and do not establish an isolated hybrid onset with a decaying low-frequency coda. All remain unconfirmed. No relative magnitude is assigned to a supposed hybrid cluster before common-source membership is established. The original three template displays remain useful observations; only the E3 example corresponds to S15C. E1/E2 displays are not S15A/B reproductions.

4 Results, validation and comparison

Extended inference produced 3,637 merged picks (560 P and 3,077 S). GaMMA provisionally associated 63 picks into 9 events. Only 3 events pass the stated geometry screen; none has completed independent VELEST/TomoDD processing. The magnitude attempt measured 140 channel amplitudes, 23 above 5 dB, and produced 0 qualifying equations and 0 calibrated magnitudes. Post-association external matching found 3 unique temporal matches within two seconds; these did not affect event detection or acceptance. The template search scanned 26 intervals (24.298 hours of valid start times), generating 4 unconfirmed candidates. Analytic association location error was 5.96e-07 km and origin-time error 0 s. A strictly downstream comparison against supplied Data S1 (54,319 events) finds 5 unique temporal matches within two seconds among 9 provisional events. For these matches, the median absolute origin-time difference is 0.350 s and median horizontal difference is 1.042 km (maximum 1.730 km). These differences describe an initial, uncalibrated sample. Temporal proximity alone does not prove event identity, and unmatched events are not declared false detections. The comparison changes no independent data or setting. The association consistency test recovers one analytically constructed homogeneous-medium event with 16 P/S arrivals. It tests association and coordinates, not earthquake-detection accuracy or the adequacy of the alternative background profile. Probability bounds, S-pick exclusions, six-pick minima and assignment uniqueness are checked. The original 27 C++ equation checks and observational waveform/FI checks are retained and rerun. All 27 checks passed. The maximum absolute FI error across the analytic mixtures was 4e-15; the maximum exact-recovery magnitude error was 1.94e-16. These errors compare against analytic fixtures, not observed earthquakes.

An additional inference comparison uses the generic original PhaseNet weights converted by SeisBench and the released retrained weights. It processes calendar-selected CSOB and CMIS observations with the same stated preprocessing. In the 1600-second CMIS window, the original model returns 17 P and 4 S picks, versus 11 P and 55 S picks from the retrained model. The waveform is dominated by persistent fluctuations. Without independently reviewed annotations, these counts establish neither

greater sensitivity nor better accuracy. The 120-second CSOB window yields no original-model pick and one retrained S pick. Neither window is claimed to match S4's illustrated example. No retraining or labeled benchmark was executed. Saved probability arrays and both plots accompany the report.

The location-consistency diagnostic compares observed and calculated S–P intervals at stations with both independently assigned phases. It does not validate location accuracy independently of the fitted model. No associated CPOZ P arrival exists in this sample, so the paper's arrival-aligned CPOZ sector gathers cannot yet be formed. Counts and residuals are saved in `data/extended/location_consistency.json`; no author pick fills this gap.

The historical leveling calculation independently extracts 128 tabulated entries from the original Del Gaudio et al. supplement, March 1905–July 2009. It subtracts the March 1905 benchmark elevation. The resulting displacement ranges from −1.266 to 2.541 m. Some source entries are estimates from photographs or earlier graphical reconstructions; their descriptions are retained. This investigation digitizes no graph. The original De Martino et al. (2021) supplement was subsequently retrieved from the publisher's static content host despite HTTP 403 on its main supplement page. Its RITE daily file supplies North, East and Up positions and uncertainties in meters, with Modified Julian Dates, through December 2019. I estimate a constant datum offset as the median difference between each overlapping leveling observation and the median GNSS Up value within 15 days. Both original series are retained; this assumes a common local displacement history and does not remove spatial differences between benchmarks. The offset and its median absolute deviation are recorded. Original daily observations for 2020–2023 remain missing. The December 2023 official monitoring bulletin gives a written approximate uplift of 1.19 m since November 2005. A separate endpoint is referenced to the median of the original November 2005 GNSS observations; no curve fills the intervening gap and no plot pixels supply values. A later interpolated compilation is excluded as a replacement.

Routine catalog pages for 2005–2025 were retrieved for explicit external context and the permitted calibration role. The current study-period download has 19,274 records, of which 12,503 have location and depth, versus 12,083 events in the publication's S5. This 420-event difference must not be removed by matching an author event list. Possible causes include archive revisions or selection differences; they are not resolved here. Context counts include unlocated events where appropriate, whereas maps require coordinates. The three-year independent detection rate, relocation precision, ring-fault geometry and magnitude-frequency distribution cannot be validated from the limited provisional sample.

After generating the displacement plot, I inspected the published Figure 1B visually. The sequence of early subsidence, uplift episodes and subsequent relaxation agrees qualitatively. The displayed vertical datum differs: the present plot subtracts the original table's March 1905 benchmark height of 3.464 m, whereas the source elevations are retained in the saved table. Absolute plot ordinates should therefore not be compared without accounting for the datum. No value was digitized or used to adjust the analysis from this comparison. The incomplete recent daily series precludes a full temporal reconstruction.

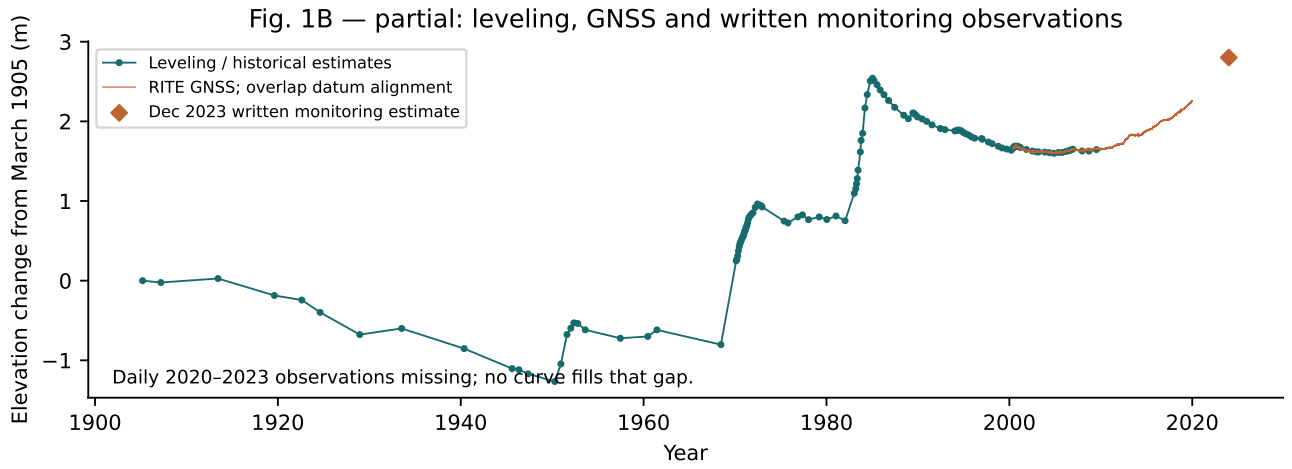


Figure 1: Figure 1B, partial. Independently extracted original leveling and GNSS observations, referenced to March 1905 using the stated overlap datum alignment. Dates denote the middle of the tabulated month; uncertain months remain flagged in CSV. Leveling ends in 2009 and daily GNSS in 2019. A separate December 2023 endpoint comes from the written monitoring summary; no curve fills the daily-observation gap.

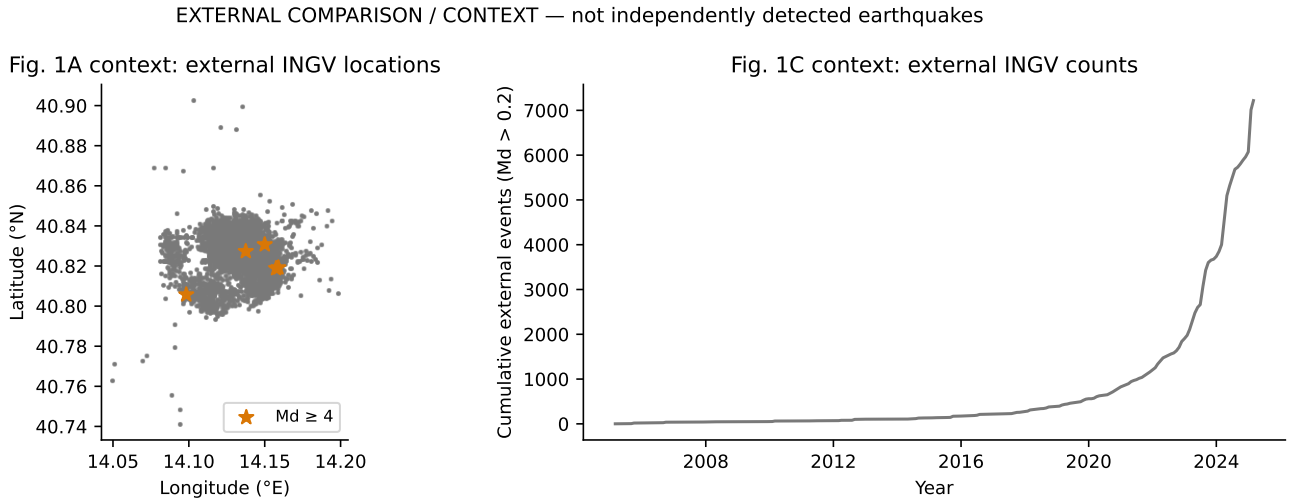


Figure 2: Clearly labeled external context related to Figure 1A/C. These are routine INGV events, not independently detected events. No external event is inserted into the independent association output.

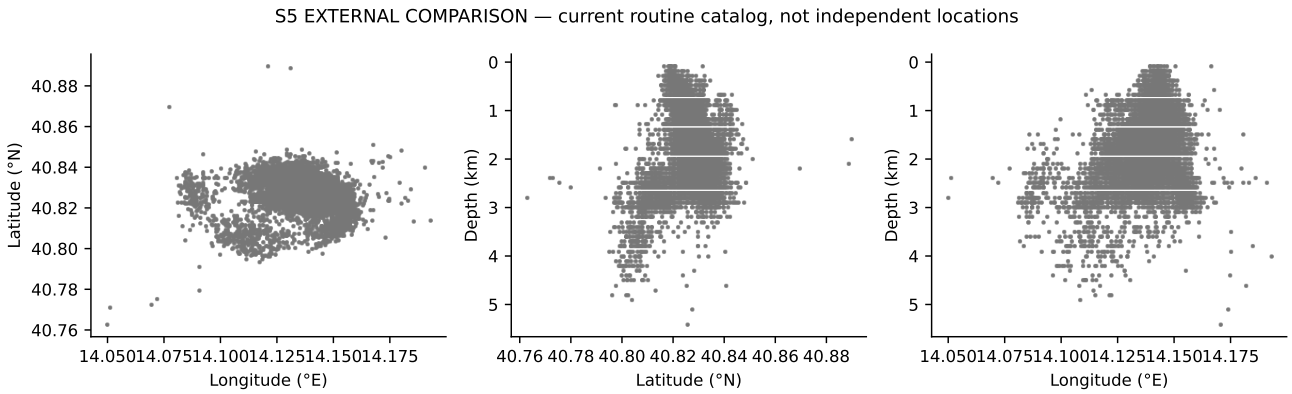


Figure 3: External comparison related to S5. Current routine-catalog locations during the study period; this is not an independent location result. The current located count differs from the paper.

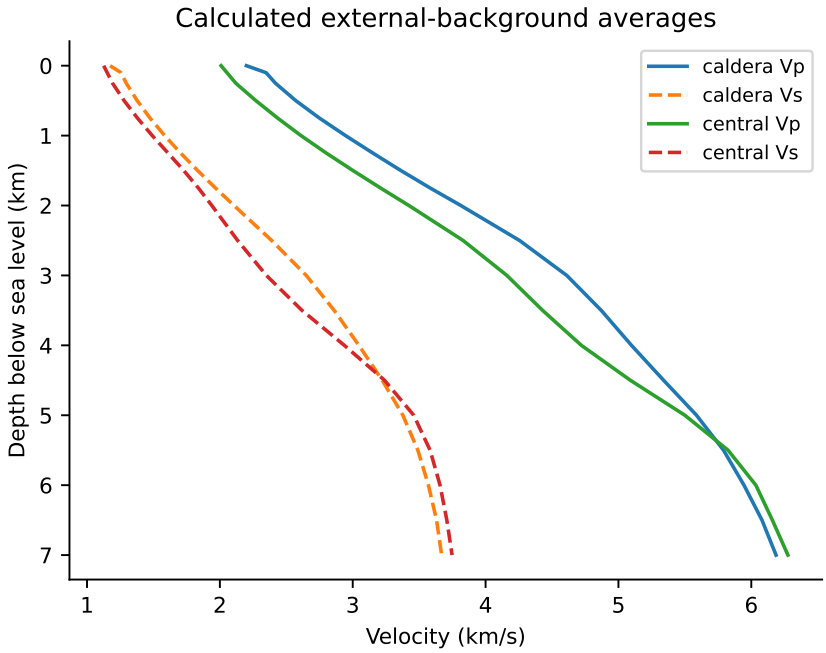


Figure 4: Independent averages of the released candidate external background. These calculated profiles are not the authors' fitted 1D model and are not asserted to equal the cited association profile.

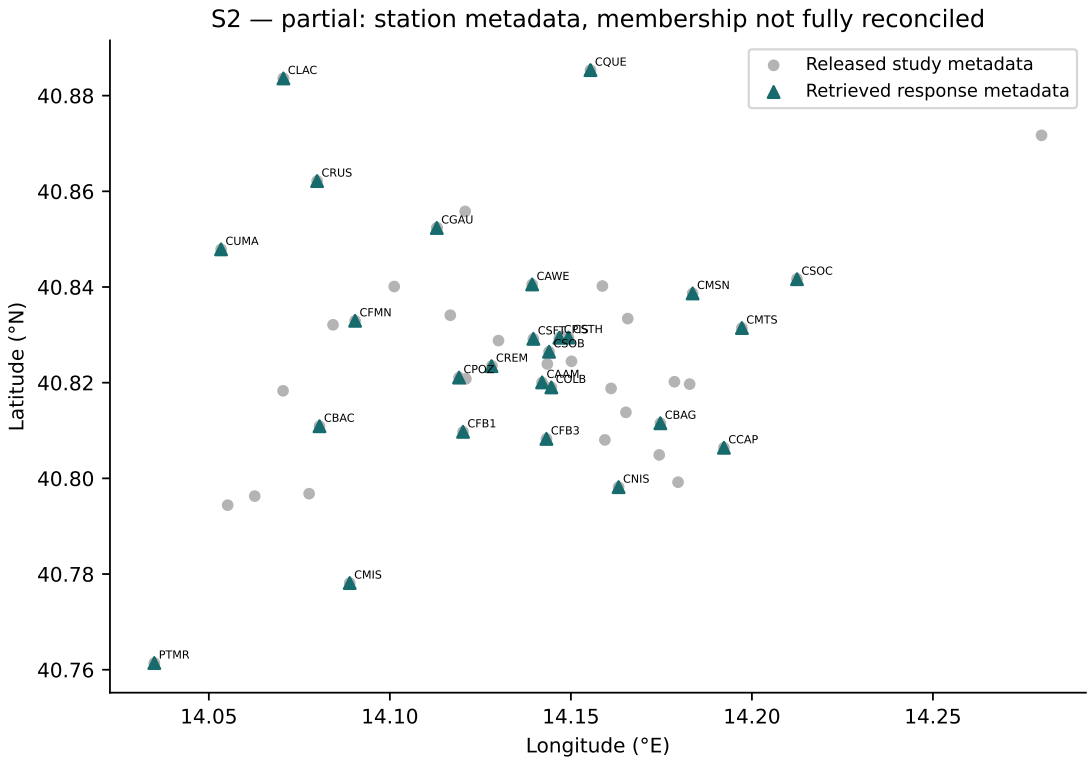


Figure 5: S2, partial station inventory. Released station coordinates and independently retrieved IV/2I response metadata. Exact 48-instrument membership, epochs and basemap are not fully reconciled.

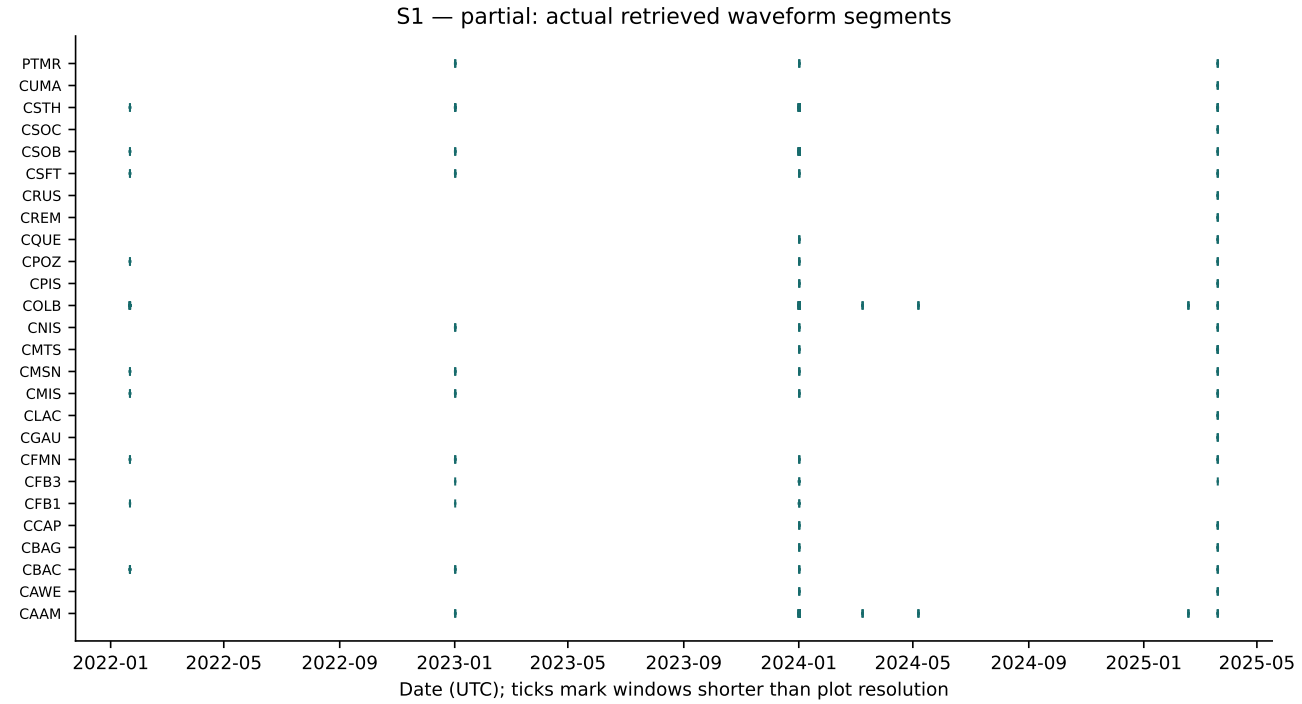


Figure 6: S1, partial actual returned intervals. Ticks identify retrieved windows shorter than plot resolution. No continuous recording between plotted dates is implied.

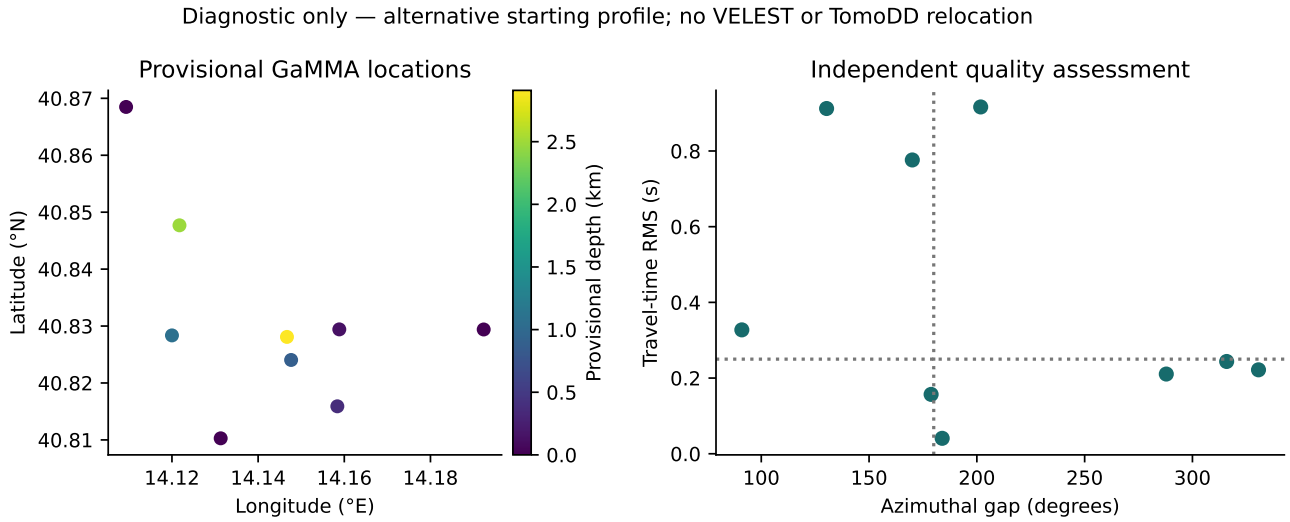


Figure 7: New diagnostic, not Figure 3. Provisional associated locations and residual/azimuthal-gap checks. The depth profile differs from the cited initial profile; neither calibrated locations nor TomoDD structures are claimed.

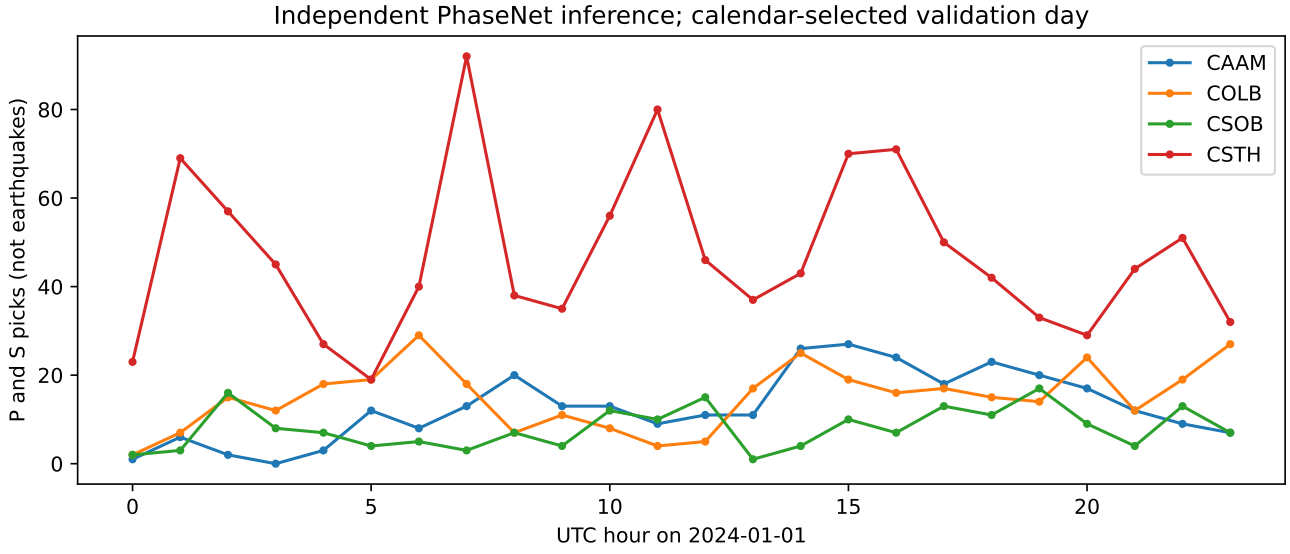


Figure 8: Retained calendar-day pick diagnostic. Model picks are not earthquake counts.

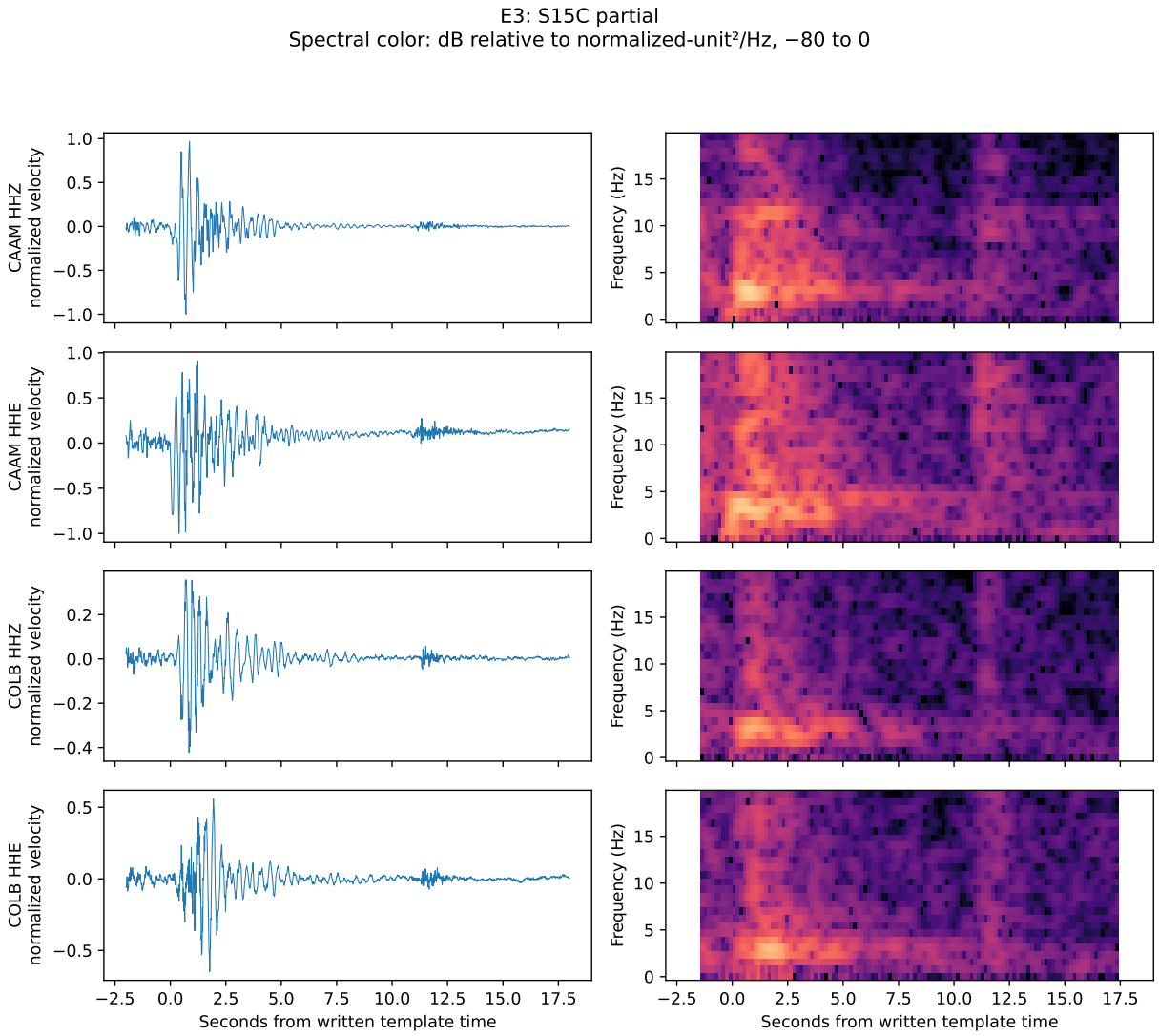


Figure 9: S15C, partial original-waveform reconstruction. Display normalization, spectral windows and sensitivity correction are explicit assumptions inherited from the retained observational implementation. It is not a source simulation.

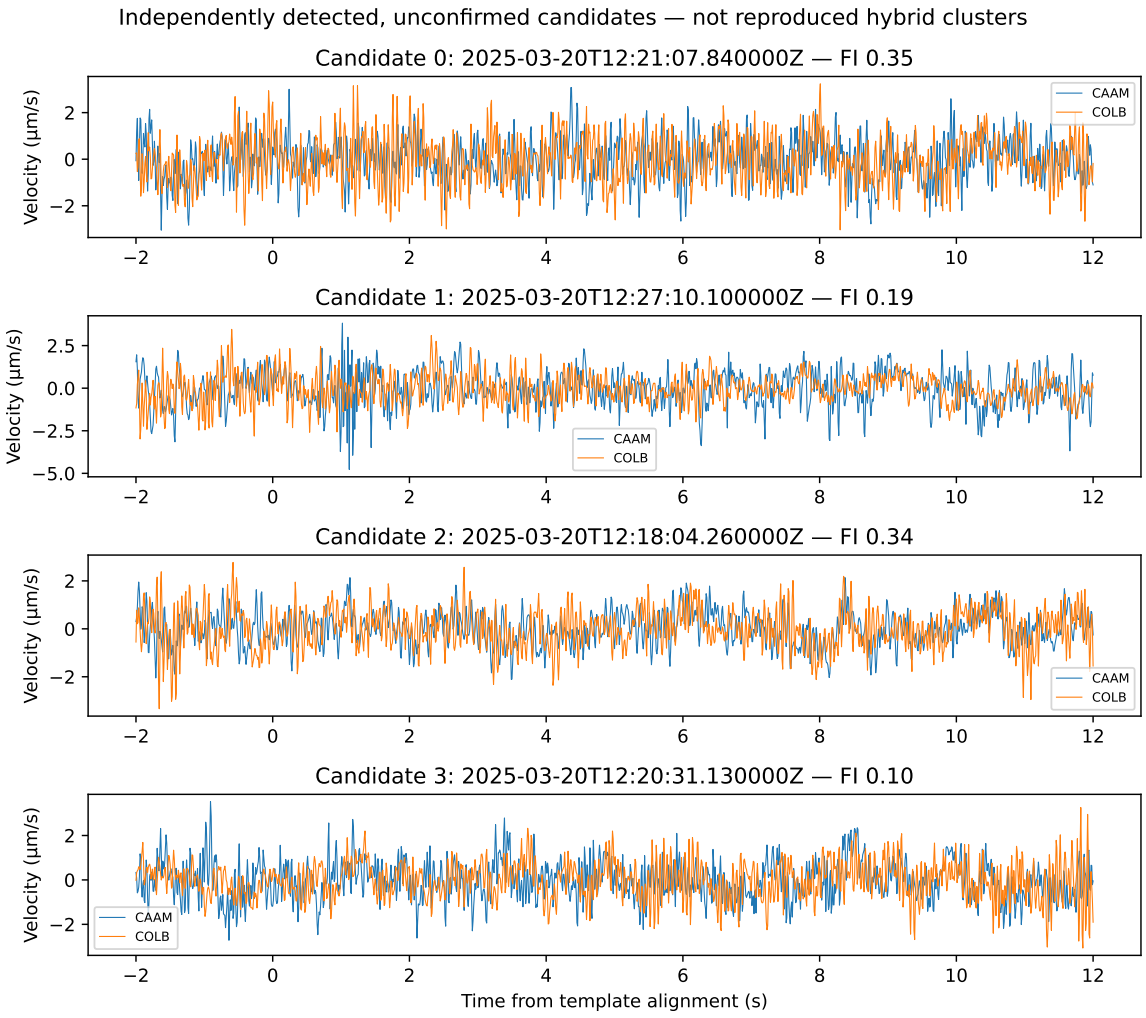


Figure 10: Independent candidate review. These correlation-threshold crossings remain unconfirmed; their appearance does not establish the paper's hybrid clusters. No supplied hybrid-event list selected these windows.

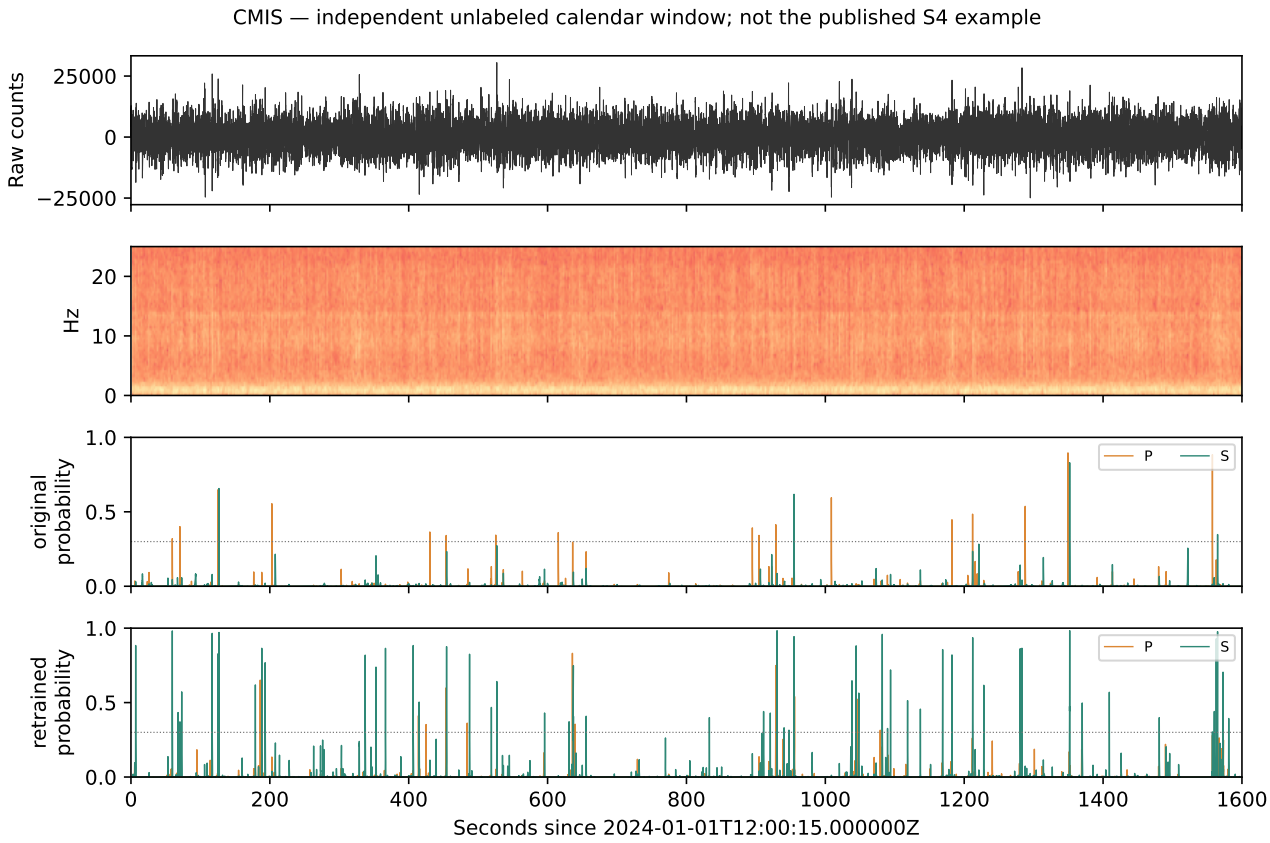


Figure 11: Additional inference comparison on original CMIS observations. The independently calculated probability outputs differ substantially. This is an unlabeled calendar window, not the published S4 example or an accuracy benchmark.

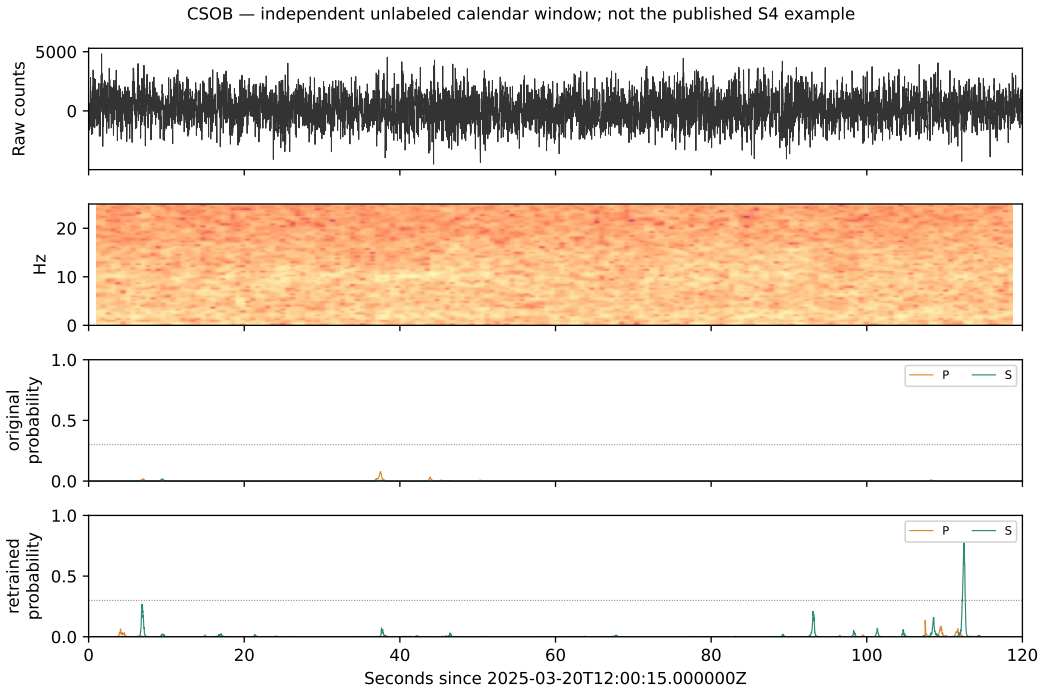


Figure 12: Additional original/retrained inference comparison at CSOB on an unlabeled calendar window. Neither the selected time nor the predicted phases is taken from an author catalog.

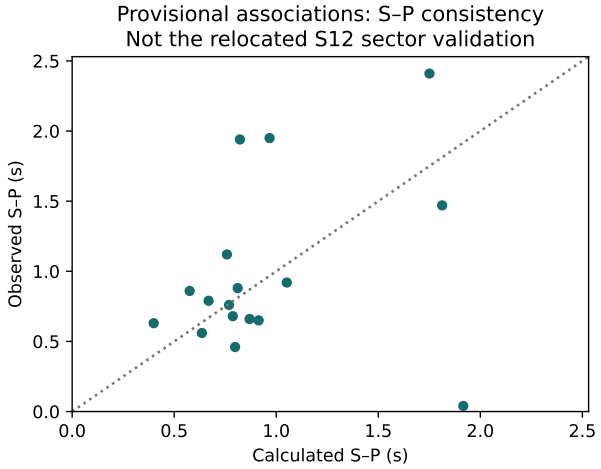


Figure 13: Internal S–P consistency of provisional associations. The dotted line denotes equality. This does not replace S12's relocated sector gathers.

5 Completion status and remaining work

No publication panel is fully independently reproduced. Figure 1B and S1, S2, S15C are partial. All remaining main-text and supplementary panels are not reproduced, including those for which an external context plot is supplied. The full record gives each quantity, written specification, exact inputs, executed data/plot commands or null commands, and a specific limitation. Generic software access is distinguished from scientific execution.

Stage	Actual execution and limitation
1. Collection	Executed calendar windows and whole-period requests; 26 stations retrieved. Full-period chunked download remains unfinished. RAN credentials unavailable.
2. Phase picking	Executed: 3637 picks; exclusions and merging checked. Full-period inference remains unfinished.
3. Training / S3–S4	Retraining not run. STEAD short links return 403; local labels and exact selections/splits missing. Original/retrained inference comparison executed on unlabeled CSOB/CMIS windows, not the exact S4 benchmark.
4. Association	Executed: 9 provisional events with an alternative external profile. Exact cited numerical starting profile missing; no author event seeds.
5. VELEST	Generic software compiled. 3 geometry-screened candidates do not resolve the chosen layered model/corrections; inversion not run.
6. TomoDD	Not run. Independent calibrated input catalog and complete differential-time/configuration workflow missing.
7. HypoDD SVD	Not run. Depends on independently relocated regional subsets and fitted model; generic software source accessible.
8. Magnitudes	Measurements executed; 0 qualifying pair equations, 0 calibrated magnitudes. Routine Md permission is resolved.
9. Completeness / b	Not calculated: no independently calibrated magnitude sample.
10. Focal mechanisms	Not run. Reviewed polarities, calibrated locations/model and gCAP configuration missing. Generic software accessible. Data S3 row-count discrepancy resolved.
11. Location checks	Residuals, gaps and internal S–P checks executed. Zero associated CPOZ P arrivals; relocated sector gathers, manual deep-event review and SVD validation unfinished.
12. Hybrid search	Executed on 24.298 available hours; 4 unconfirmed candidates. Three-year classification, clustering and catalog-wide FI review unfinished.
13. Historical context	Executed original leveling-table extraction (128 entries) and external routine-catalog context. Original RITE GNSS processed through 2019 with explicit overlap alignment; daily 2020–2023 continuation missing.

The unresolved starting profile, missing acceleration access and incomplete full-period processing are distinct limitations. Lack of a full catalog is not attributed solely to the prompt. Full retrieval, qualified calibration-event selection, VELEST inversion, coherency-weighted differential times, TomoDD relocation, HypoDD uncertainty, reviewed polarities, SKHASH and gCAP inversion, complete magnitude statistics and multiyear hybrid classification remain necessary. Unknown settings require documented

assumptions and sensitivity checks rather than hidden substitutions. No coupled physical model is invented.

6 Rebuild, provenance and replacement

The README gives environment creation, exact snapshot restoration and the clean rebuild command. The offline rebuild verifies supplied-source and downloaded-input checksums before removing generated products. It reruns the original checks and waveform processing, extended picking/association, analytic association validation, external context extraction, response-corrected amplitude/quality measurements, template scans, plotting, panel tables and two LaTeX passes. Command outcomes are saved in `data/build_log.json`. Input hashes and placement URLs are in `docs/rebuild_inputs.json`. Live routine pages can change; restoration refuses a mismatched hash and asks for the recorded snapshot or an explicit new provenance audit. Successful HTTP access is not taken as proof of reproducibility of an older mutable archive version.

The previous report source/PDF, station/coverage figures, panel records, input policy, blocker descriptions and rebuild procedure are replaced. Prior observational template figures, raw source documents, independent C++ checks and useful setup are preserved; historical erroneous products remain only in Git history. New outputs are saved under `data/extended/`. Exact replacement details are in `docs/REPLACEMENTS.md`. This bounded rebuild does not declare the scientific project complete.

All project inputs, environments and intermediate products reside under this repository in `inputs/`, `.venv/`, `tmp/` and `build/`. Standard dependency caches may reside in `/home/mehffin/.cache/pip`; the earlier initial SeisBench configuration remains in `/home/mehffin/.seisbench`. Current analysis redirects SeisBench, plotting and numba caches into the repository. Large inputs and generic software downloads are not committed. Their URLs, sizes, checksums or pinned revisions accompany the project.

A Figure-by-figure record

The following table records every publication panel or labeled view. For every panel marked not reproduced, no command generates a claimed reproduction; null command fields in the machine-readable record mean unexecuted, not unavailable software. The complete per-panel input paths and command fields are in docs/panel_record.json. Each listed external file's exact URL, size and checksum appears in docs/rebuild_inputs.json. The partial panels use the recipes below, from the repository root. Each script is invoked with .venv/bin/pythonscripts/NAME.py; required upstream outputs are produced by scripts/rebuild.py--clean.

- 1B: data context_data.py; plot plot_extended.py. Inputs: inputs/extended/delgaudio2010_table.doc, inputs/extended/demartino_supp.zip (member RITE_daily.txt), and inputs/extended/monthly_2023_12.pdf.
- S1: data analyze_observations.py and extended_analysis.py; plot plot_extended.py. Inputs: all 171 miniSEED paths explicitly listed in the S1 record and frozen manifest. Actual recorded segments supply the plotted intervals.
- S2: data analyze_observations.py and extended_analysis.py; plot plot_extended.py. Inputs: inputs/models/stations.csv, inputs/metadata/channels.txt, and the 26 XML response files explicitly listed in the S2 record under inputs/extended/responses/.
- S15C: data analyze_observations.py; plot plot_observations.py. Inputs: inputs/waveforms/E3_CAAM.mseed, inputs/waveforms/E3_COLB.mseed, and inputs/metadata/response.xml.

External context recipes for 1A/C and S5 use context_data.py and plot_extended.py; the input routine HTML pages are identified row by row in the saved external CSV. Additional diagnostics use compare_pickers.py, location_consistency.py, or review_hybrid_candidates.py, with saved numerical outputs. These commands do not upgrade the corresponding unreproduced publication panels.

Panel/view	Quantity	Status, specification and limitation
1A	INGV event map and large-event markers	not reproduced. Main p.71, Fig.1 caption. 2005-01-01 to 2025-03-20; origin 14.14 E, 40.82 N; sea-level depth; Md>=4 markers. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
1B	Vertical displacement at Pozzuoli port	partially reproduced. Main p.71, Fig.1 caption. 1905-2023; Rione Terra. Original leveling table independently extracted and referenced to March 1905; 128 entries through July 2009. Original RITE GNSS data through 2019 are independently aligned using overlapping leveling epochs; 2020–2023 daily observations remain missing.
1C	Temporal seismicity and cumulative counts	not reproduced. Main p.71, Fig.1 caption. 2005-2025; cumulative Md>0.2; study interval 2022-01-21 to 2025-03-20. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
2A	Catalog seismicity rates by region	not reproduced. Main p.72, Fig.2 caption. Central vs ring/other regions; 54319 vs 12083 catalog events; time bin unspecified. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.

Panel/view	Quantity	Status, specification and limitation
2B	Depth versus time, symbol size by magnitude	not reproduced. Main p.72, Fig.2 caption. Sea-level depth; near-surface 0-0.5 km; central 0.5-2.1; ring sectors to 3.7 km. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
3/map	Epicentral spatial pattern	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
3/northing-depth	Depth versus northing	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
3/ easting-depth	Depth versus easting	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
4A	Ring-sector focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Independent calibrated locations, reviewed polarities and gCAP configurations are missing; focal inversions not run. Data S3 actually contains 74 rows/58 gCAP entries, consistent with methods; caption says 55.
4B	Central/onshore focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Independent calibrated locations, reviewed polarities and gCAP configurations are missing; focal inversions not run. Data S3 actually contains 74 rows/58 gCAP entries, consistent with methods; caption says 55.
S1	Station data-availability timeline	partially reproduced. Supplement S12 caption; S2 methods. 48 stations; continuous-data availability. Verified retrieved channel intervals only; full-period network collection remains unfinished and RAN credentials are unavailable.
S2	Station map	partially reproduced. Supplement S13 caption; S2 methods. 23 broadband, 9 short-period, 16 accelerometers. 51-station permitted metadata inventory differs from the paper's 48-station description; exact epoch/membership and basemap not reproduced.
S3A	Noise example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3B	Single-event example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.

Panel/view	Quantity	Status, specification and limitation
S3C	Two overlapping events: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3D	Five overlapping copies of one event: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S4A	Swarms at CSOB: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 120 s; ~2 h after May 2024 Md4.4 event; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark.
S4B	CMIS noise: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 1600 s; coastal noise; phase threshold 0.3. Original and retrained models both executed on independently selected CSOB/CMIS calendar windows. Exact published windows and independent reference annotations remain unidentified; these diagnostics are not the published benchmark.
S5/map	Routine-catalog spatial views; map	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
S5/northing-depth	Routine-catalog spatial views; northing-depth	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
S5/ easting-depth	Routine-catalog spatial views; easting-depth	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. External routine-catalog context is plotted and labeled separately. It is not an independently detected historical catalog or a reproduced analysis result.
S6/map	Initial-location spatial views; map	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. Generic VELEST compiled. The small independent candidate set is insufficient to resolve the chosen layered velocities and station corrections; paper uses 373 calibration events. Inversion not executed.
S6/northing-depth	Initial-location spatial views; northing-depth	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. Generic VELEST compiled. The small independent candidate set is insufficient to resolve the chosen layered velocities and station corrections; paper uses 373 calibration events. Inversion not executed.
S6/ easting-depth	Initial-location spatial views; easting-depth	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. Generic VELEST compiled. The small independent candidate set is insufficient to resolve the chosen layered velocities and station corrections; paper uses 373 calibration events. Inversion not executed.
S7A/map	Cluster locations and 2-sigma errors: map and cross-sectional views; map	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.

Panel/view	Quantity	Status, specification and limitation
S7A/ easting-depth	Cluster locations and 2-sigma errors: map and cross-sectional views; easting-depth	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7B	EX uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7D	EY uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7F	EZ uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7C	EX uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7E	EY uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S7G	EZ uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent relocated catalog or fitted 1D model for regional HypoDD SVD tests. Generic HypoDD source is accessible; scientific runs are unfinished.
S8A	Estimated local versus routine duration magnitude	not reproduced. Supplement S19 caption; S5-S6 methods. Relative-amplitude LS; SNR>=5 dB; >=2 common stations; separation<distance/5. Independent amplitude/SNR measurements executed but no qualifying relative-magnitude edge; zero calibrated magnitudes. INGV Md is now a permitted required calibration reference; no catalog magnitudes fill missing results.
S8B	Magnitude-frequency distribution and fitted b value	not reproduced. Supplement S19 caption; S6 methods. Published Mc=0.4 by maximum curvature; b=0.97 by ML; bin size unspecified. Independent amplitude/SNR measurements executed but no qualifying relative-magnitude edge; zero calibrated magnitudes. INGV Md is now a permitted required calibration reference; no catalog magnitudes fill missing results.
S9/map	Filtered-catalog spatial views and feature labels A-D; map	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S9/northing-depth	Filtered-catalog spatial views and feature labels A-D; northing-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.

Panel/view	Quantity	Status, specification and limitation
S9/ easting-depth	Filtered-catalog spatial views and feature labels A-D; easting-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2022	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2022. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2023	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2023. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2024	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2024. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S10/2025	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2025 through March20. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-1	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-2	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-3	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S11/swarm-4	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S12A	Ring-sector division	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.

Panel/view	Quantity	Status, specification and limitation
S12B	Ring-sector division in complementary spatial view	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12C	Common-receiver gather: east / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12D	Common-receiver gather: southwest / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12E	Common-receiver gather: north / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12F	Common-receiver gather: east / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12G	Common-receiver gather: southwest / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S12H	Common-receiver gather: north / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. CPOZ observations retrieved, but this provisional catalog contains zero independently associated CPOZ P arrivals. Internal S-P consistency calculated at other stations; relocated sector gathers and location validation remain unfinished.
S13A	Full caldera seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.

Panel/view	Quantity	Status, specification and limitation
S13B	Onshore seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Unfinished full-period processing and missing calibrated/relocated catalog. Provisional GaMMA diagnostics use an alternative external profile; they are not this panel. Author-derived results are not substituted.
S14/1	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S14/2	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S14/3	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S14/ unclustered	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. All supported downloaded CAAM/COLB intervals were scanned; exposure is about one day, not three years. Four candidate review examples remain unconfirmed. Published cluster histories and cluster-relative magnitudes are not reproduced.
S15A	Hybrid family 1: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.
S15B	Hybrid family 2: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.
S15C	Hybrid family 3: waveforms and spectrograms	partially reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Written template times used; exact published example alignment, display filtering and spectral settings incompletely specified. Sensitivity-corrected velocities and stated independent display settings used.
S15D	Shallow volcano-tectonic example: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Exact shallow VT example not identified by written method alone; no author catalog or plot values used to select it.

B Equation consistency checks

Check	Absolute error	Tolerance
equal_amplitudes	0	1e-14
decade_amplitude	0	1e-14
ratio_reversal	0	1e-14
common_station_gain	5.55e-17	1e-14
zero_amplitude_rejected	0	0
negative_power_rejected	0	0
fi_mixture_0	4e-15	1e-10
fi_mixture_1	8.88e-16	1e-10
fi_mixture_2	1.78e-15	1e-10
fi_mixture_3	1.55e-15	1e-10
fi_mixture_4	1.78e-15	1e-10
fi_mixture_5	2.11e-15	1e-10
fi_mixture_6	2.22e-16	1e-10
fi_mixture_7	2.11e-15	1e-10
fi_mixture_8	1.33e-15	1e-10
fi_mixture_9	8.88e-16	1e-10
fi_mixture_10	1.33e-15	1e-10
fi_mixture_11	4.44e-16	1e-10
fi_mixture_12	1.78e-15	1e-10
parseval	3.33e-15	1e-10
fi_circular_shift	7.22e-16	1e-10
fi_gain	8.88e-16	1e-10
snr_5s_2s_windows	8.88e-16	1e-10
magnitude_exact_recovery	1.94e-16	1e-12
magnitude_pair_order	0	1e-12
magnitude_gauge_shift	4.44e-16	1e-12
unanchored_problem_rejected	0	0

C Sources

Primary specification: Tan et al. (2025), Science 390, 70–75, local `tanmodel.pdf` and `tanmodelsupp.pdf`. Original waveform and response source: https://webservices.ingv.it/fdsnws/. Required routine magnitude reference and external context: https://terremoti.ov.ingv.it/gossip/flegrei/. Original leveling table: Del Gaudio et al. (2010), doi:10.1016/j.jvolgeores.2010.05.014, publisher supplement `1-s2.0-S0377027310001745-mmc1.doc`. External background references: Battaglia et al. (2008), doi:10.1111/j.1365-2478.2007.00687.x; Calò and Tramelli (2018), doi:10.1038/s41598-018-34456-x. Trained artifact and candidate fixed background release: https://github.com/iamharrison2000/Campi-Flegrei-Earthquake-Monitoring, revision pinned in the receipts. GaMMA: https://github.com/AI4EPS/GaMMA, revision `80394dd4a4c29450da599f507916ecf1db6203dd`. VELEST: https://seg.ethz.ch/software/velest.html. Additional source receipts record RAN, temporary observations, STEAD, HypoDD, SKHASH and gCAP. No Tan study-specific analysis or plotting script was used.